Having standardized our dataset, (10.59) only yields the projections in the context of the standardized dataset. To obtain our projection in the original data space (i.e., before standardization), we need to undo the standardization (10.58) and multiply by the standard deviation before adding the mean so that we obtain

$$\tilde{x}_*^{(d)} \leftarrow \tilde{x}_*^{(d)}\sigma_d + \mu_d\,, \quad d = 1, \ldots, D\,. \tag{10.61}$$

Figure 10.11(f) illustrates the projection in the original data space.

Example 10.4 (MNIST Digits: Reconstruction)
In the following, we will apply PCA to the MNIST digits dataset, which contains $60{,}000$ examples of handwritten digits 0 through 9. Each digit is an image of size 28×28, i.e., it contains 784 pixels so that we can interpret every image in this dataset as a vector $\boldsymbol{x} \in \mathbb{R}^{784}$. Examples of these digits are shown in Figure 10.3.

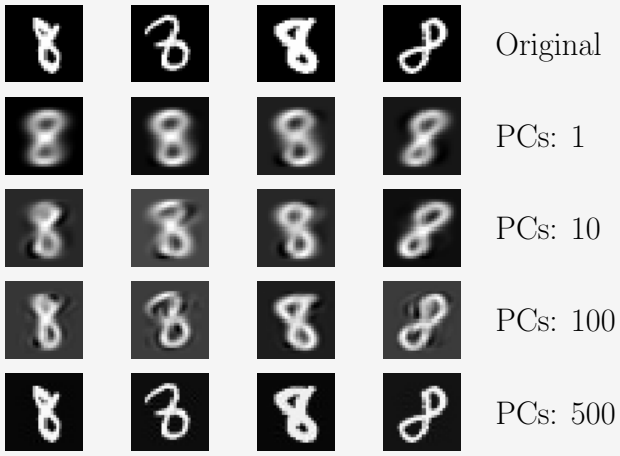


Figure 10.12 Effect of increasing the number of principal components on reconstruction.

For illustration purposes, we apply PCA to a subset of the MNIST digits, and we focus on the digit "8". We used 5,389 training images of the digit "8" and determined the principal subspace as detailed in this chapter. We then used the learned projection matrix to reconstruct a set of test images, which is illustrated in Figure 10.12. The first row of Figure 10.12 shows a set of four original digits from the test set. The following rows show reconstructions of exactly these digits when using a principal subspace of dimensions 1, 10, 100, and 500, respectively. We see that even with a single-dimensional principal subspace we get a halfway decent reconstruction of the original digits, which, however, is blurry and generic. With an increasing number of principal components (PCs), the reconstructions become sharper and more details are accounted for. With 500 prin-

cipal components, we effectively obtain a near-perfect reconstruction. If we were to choose 784 PCs, we would recover the exact digit without any compression loss.

Figure 10.13 shows the average squared reconstruction error, which is

$$\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2 = \sum_{i=M+1}^{D}\lambda_i\,, \tag{10.62}$$

as a function of the number M of principal components. We can see that the importance of the principal components drops off rapidly, and only marginal gains can be achieved by adding more PCs. This matches exactly our observation in Figure 10.5, where we discovered that the most of the variance of the projected data is captured by only a few principal components. With about 550 PCs, we can essentially fully reconstruct the training data that contains the digit "8" (some pixels around the boundaries show no variation across the dataset as they are always black).

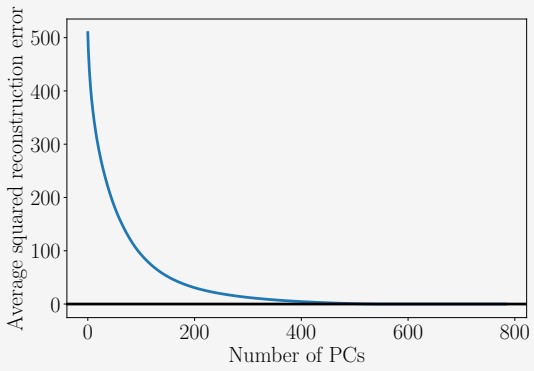


Figure 10.13 Average squared reconstruction error as a function of the number of principal components. The average squared reconstruction error is the sum of the eigenvalues in the orthogonal complement of the principal subspace.

10.7 Latent Variable Perspective

In the previous sections, we derived PCA without any notion of a probabilistic model using the maximum-variance and the projection perspectives. On the one hand, this approach may be appealing as it allows us to sidestep all the mathematical difficulties that come with probability theory, but on the other hand, a probabilistic model would offer us more flexibility and useful insights. More specifically, a probabilistic model would

- Come with a likelihood function, and we can explicitly deal with noisy observations (which we did not even discuss earlier)
- Allow us to do Bayesian model comparison via the marginal likelihood as discussed in Section 8.6
- View PCA as a generative model, which allows us to simulate new data

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